Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Integration

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find $\int x^3 \, dx$. **[2]**

2. Find $\int (4x^3 - 2x + 5) \, dx$. **[3]**

3. Find $\int \frac{1}{x^2} \, dx$. **[2]**

4. Find $\int \sqrt{x} \, dx$. **[2]**

5. Find $\int (3x^{1/2} - x^{-1/2}) \, dx$. **[3]**

6. Find $\int_0^2 (3x^2 - 1) \, dx$. **[3]**

7. Find $\int_1^4 \sqrt{x} \, dx$. **[3]**

8. Find $\int_0^1 (2x + 3) \, dx$. **[3]**

9. Find the area under $y = x^2$ from $x = 0$ to $x = 3$. **[3]**

10. Find the area under $y = 4 - x^2$ from $x = 0$ to $x = 2$. **[4]**

11. Find $\int (x + 1)^2 \, dx$. **[3]**

12. Find $\int \frac{2}{x^3}\,\mathrm{d}x$. **[2]**

13. Given $\frac{dy}{dx} = 3x^2 - 2x$ and $y = 1$ when $x = 0$, find y. **[4]**

14. Given $\frac{dy}{dx} = 2x + 1$ and $y = 4$ when $x = 1$, find y. **[4]**

15. Find the area bounded by $y = x(x - 2)$ and the x-axis. **[5]**

Section B – Hard (Q16–25)

16. Find the area between $y = x^2$ and $y = x + 2$. **[6]**

17. Find the area between $y = x^2 - 1$ and $y = 3$. **[5]**

18. Find $\int_0^{\pi/2} \sin x\,\mathrm{d}x$. **[3]**

19. Find $\int_0^{\pi/4} \cos 2x\,\mathrm{d}x$. **[3]**

20. Find the area under $y = \sin x$ from $x = 0$ to $x = \pi$. **[3]**

21. Given $\frac{dy}{dx} = 6x - 4$ and the curve passes through $(1, 5)$, find the equation of the curve. **[4]**

22. Find the area bounded by $y = 4x - x^2$ and the x-axis. **[5]**

23. Evaluate $\int_1^2 \left(x^2 + \frac{1}{x^2}\right) \mathrm{d}x$. **[4]**

24. Find $\int (2x - 1)^3 \, \mathrm{d}x$. **[4]**

25. The curve $y = f(x)$ has gradient $3x^2 - 6x$ and passes through the origin. Find the area under the curve from $x = 0$ to $x = 3$. **[6]**

End of Worksheet • *Show all working clearly.*